

Craniodiaphyseal Dysplasia (CDD): A Comprehensive Disease Characterization Report

Target disease: Craniodiaphyseal Dysplasia **MONDO:** MONDO:0021021 | **OMIM:** 122860 (autosomal dominant) | **Orphanet:** ORPHA:1513 | **Category:** Mendelian sclerosing bone dysplasia

Summary

Craniodiaphyseal dysplasia (CDD) is an extremely rare, severe, progressive sclerosing bone dysplasia characterized by massive generalized hyperostosis and sclerosis of the skull and facial bones ("leontiasis ossea") together with diaphyseal hyperostosis and defective modeling of the long bones. The relentless deposition of bone progressively narrows the craniofacial foramina and encroaches on the brain, compressing cranial nerves, the foramen magnum, and intracranial contents. The clinical consequences — blindness, sensorineural and conductive hearing loss, facial palsy, raised intracranial pressure, seizures, and premature death — arise directly from this mechanical encroachment rather than from a metabolic or systemic derangement.

The molecular basis of the classic **autosomal-dominant** form is a defect in the secretion of **sclerostin**, the osteocyte-derived protein encoded by *SOST*. Heterozygous mutations in the *SOST* secretion signal peptide (**c.61G>A**, **p.Val21Met**; **c.61G>T**, **p.Val21Leu**) greatly reduce sclerostin secretion through a **dominant-negative** mechanism, which distinguishes CDD from the recessive loss-of-function *SOST* disorders sclerosteosis and van Buchem disease. Sclerostin normally binds the Wnt co-receptors LRP4/5/6 to antagonize canonical Wnt/ β -catenin signaling in osteoblasts; when secreted sclerostin is reduced, this brake on bone formation is released, canonical Wnt signaling is de-repressed, and osteoblast-driven hyperostosis ensues. A phenotypically overlapping **recessive** form is caused by biallelic loss-of-function variants in *SP7/Osterix*, an osteoblast master transcription factor, establishing genetic heterogeneity for the CDD phenotype.

There is no approved disease-modifying pharmacotherapy. Management is symptomatic and surgical — decompressive craniectomy, staged craniofacial reduction osteoplasty, and rehabilitation of hearing loss (including bone-anchored hearing aids). The sclerostin–Wnt axis is strongly validated in the opposite (therapeutic) direction by the anti-sclerostin antibody romosozumab, which is bone-anabolic in osteoporosis — confirming the direction of CDD causality but implying that no safe pro-sclerostin ("bone-quieting") therapy yet exists. Preclinical leads include the chemical chaperone sodium 4-phenylbutyrate (4-PBA), which suppresses hyperostosis in an osteocyte unfolded-protein-response (UPR) mouse model consistent with CDD, and competitive modulation of the LRP4/6–sclerostin interface.

1. Disease Information

Overview. CDD is "a rare, sporadic form of craniotubular bone dysplasia, characterized by massive generalized hyperostosis and sclerosis, particularly of the skull and facial bones, leading to severe deformity" ([PMID: 8827383](#)). It is regarded as the most severe member of the *SOST*-related craniotubular hyperostosis spectrum.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0021021
OMIM	122860 (Craniodiaphyseal dysplasia, autosomal dominant)
Orphanet	ORPHA:1513 (Craniodiaphyseal dysplasia)
ICD-10	Q78.8 (Other specified osteochondrodysplasias)
ICD-11	LD24.Y / FB80.Y (skeletal dysplasia grouping)
MeSH	Craniofacial / hyperostosis terms (no dedicated unique descriptor)
Gene(s)	<i>SOST</i> (dominant); <i>SP7</i> /Osterix (recessive)

Synonyms / alternative names. Craniodiaphyseal dysplasia; CDD; "leontiasis ossea" is a descriptive term for the facial appearance (not a synonym for the disease as a whole).

Information source. The evidence base is derived almost entirely from **aggregated disease-level resources and individual published case reports / small case series** (OMIM, Orphanet, and a limited primary literature of fewer than ~two dozen reported patients), rather than from EHR-derived cohorts. This is a direct consequence of the disease's extreme rarity.

2. Etiology

Primary cause — genetic. CDD is a monogenic Mendelian disorder.

- **Autosomal-dominant CDD** is caused by heterozygous missense mutations in the **secretion signal peptide of *SOST*** (sclerostin). Kim et al. (2011) identified c.61G>A (p.Val21Met) and c.61G>T (p.Val21Leu) in two unrelated children with CDD: *"We discovered mutations c.61G>A (Val21Met) and c.61G>T (Val21Leu) [in] two children with CDD. As these mutations are located in the secretion signal of the SOST gene, we tested their effect on secretion by transfecting the mutant constructs into 293E cells. Intriguingly, these mutations greatly reduced the secretion of SOST"* (PMID: 21221996).
- **Autosomal-recessive CDD** is caused by biallelic variants in ***SP7/Osterix***, an osteoblast transcription factor (Hendrickx et al. 2023, PMID: 36436818; Gauthier et al. 2024, PMID: 37918503).

Genetic risk factors. The causal variants are themselves the risk determinants; there are no established common susceptibility loci or modifier genes for this ultra-rare Mendelian condition. Most dominant cases appear to arise **de novo** (sporadic), consistent with the disease's severity and reproductive impact.

Environmental risk factors. None established. CDD is not attributable to toxins, infection, nutrition, occupation, age, or sex. Family history is relevant only in the rare familial / recessive settings.

Protective factors. None identified. No protective alleles or environmental exposures are known.

Gene–environment interactions. None documented. The osteocyte UPR (endoplasmic reticulum stress) has been proposed as an intracellular **contributory amplifier** of the phenotype (see Section 6), but this is a cell-biological mechanism rather than an environmental exposure.

3. Phenotypes

CDD phenotypes are predominantly **physical/skeletal manifestations and clinical signs**, with secondary **neurological** signs from bony encroachment. Onset is in **early childhood**, severity is **severe**, and the course is **progressive**.

Phenotype	Type	Suggested HPO	Onset / severity / course	Notes & evidence
Cranial hyperostosis / sclerosis	Physical/imaging	HP:0004493 (Thickened calvaria); HP:0004437 (Cranial hyperostosis)	Childhood; severe; progressive	Calvarial thickness "nearly 4 cm" (PMID: 8827383)
Facial hyperostosis / distortion ("leontiasis ossea")	Physical sign	HP:0011856 (facial bone hyperostosis)	Childhood; severe; progressive	Massive facial bone deposition (PMID: 8827383)
Macrocephaly	Physical sign	HP:0000256	Childhood	(PMID: 14564212)
Diaphyseal hyperostosis / defective long-bone modeling (undertubulation)	Imaging	HP:0100670 (Diaphyseal sclerosis); HP:0005791 (Undertubulation)	Childhood; progressive	(PMID: 1987972; PMID: 14564212)
Clavicular / rib / axial sclerosis	Imaging	HP:0100692 (Sclerotic clavicle)	Childhood; progressive	(PMID: 14564212)
Optic atrophy / visual loss / blindness	Neurological sign	HP:0000648 (Optic atrophy); HP:0000618 (Blindness)	Childhood–adolescence; progressive	Cranial-nerve II compression (PMID: 8827383)
Hearing loss (sensorineural + conductive)	Sign	HP:0000407 (SNHL); HP:0000405 (conductive)	Childhood; progressive	(PMID: 8827383; PMID: 31132523)
Facial nerve palsy	Sign	HP:0010628 (Facial palsy)	Childhood; progressive	Foraminal narrowing (PMID: 8827383)

Phenotype	Type	Suggested HPO	Onset / severity / course	Notes & evidence
Raised intracranial pressure	Sign	HP:0002516	Childhood; progressive	(PMID: 8827383; PMID: 1987972)
Seizures	Sign	HP:0001250	Variable	(PMID: 1987972)

Frequency. Because reported cases number only in the low tens, phenotype frequencies are qualitative. Craniofacial hyperostosis, facial distortion, and diaphyseal long-bone involvement are essentially **universal (defining features)**; cranial-nerve compression syndromes (visual loss, hearing loss, facial palsy) and raised ICP are **common** and progressive.

Quality-of-life impact. Severe and multi-domain: progressive sensory loss (blindness, deafness), facial disfigurement, chronic headache from raised ICP, and neurological morbidity substantially impair daily functioning. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease.

4. Genetic / Molecular Information

Causal genes.

Gene	HGNC / locus	Inheritance	OMIM	Role
<i>SOST</i> (sclerostin)	HGNC:13771; 17q21.31	Autosomal dominant	122860 (CDD); 605740 (<i>SOST</i>)	Osteocyte-secreted Wnt antagonist
<i>SP7</i> / Osterix	HGNC:17321; 12q13.13	Autosomal recessive	606633 (<i>SP7</i>)	Osteoblast master transcription factor

Pathogenic variants (dominant *SOST*).

- **c.61G>A (p.Val21Met)** and **c.61G>T (p.Val21Leu)** — missense substitutions within the **secretion signal peptide**. Both were shown to greatly reduce *SOST* secretion in transfected 293E cells (PMID: 21221996).

- **Variant type/class:** missense, signal-peptide.
- **Functional consequence: dominant-negative** reduction of extracellular sclerostin (not simple haploinsufficiency). Kim et al. conclude: *"Unlike the other SOST-related conditions, sclerosteosis and Van Buchem disease that are inherited as recessive traits[, CDD] seem to be caused by a dominant negative mechanism"* (PMID: 21221996).
- **Classification (ACMG/AMP):** pathogenic (functional secretion assay + segregation with severe phenotype).
- **Allele frequency:** absent from population databases (gnomAD) — private, disease-causing variants.
- **Origin:** germline; dominant cases frequently de novo.

Pathogenic variants (recessive SP7). Gauthier et al. reported biallelic loss-of-function variants — **c.359_362del (p.Asp120Valfs*11)** and **c.1163_1174delinsT (p.Pro388Leufs*33)** — producing a sclerotic skeletal dysplasia overlapping juvenile Paget's disease and CDD: *"SP7 variants may also cause sclerotic skeletal dysplasias (SSD), partially overlapping with Juvenile Paget's disease and craniodiaphyseal dysplasia, characterized by skull hyperostosis, long bones sclerosis, large ribs and clavicles, and possible recurrent fractures"* (PMID: 37918503). These are biallelic frameshift loss-of-function variants.

Modifier genes. None formally established. The wider Wnt/sclerostin axis genes (*LRP4*, *LRP5*, *LRP6*) are mechanistically adjacent (Section 6) and were explicitly excluded in a mild adult CDD-like case (Janssens et al. 2003, PMID: 14564212).

Epigenetic information. No disease-specific DNA-methylation or histone data for CDD. At the mechanistic level, osteocyte ER stress/UPR modulates *SOST* transcription (Section 6), but this is regulatory rather than a documented epigenetic mark.

Chromosomal abnormalities. None; CDD is a single-gene disorder without characteristic aneuploidy, translocation, or copy-number signature.

5. Environmental Information

- **Environmental factors:** None identified. CDD is not linked to toxins, radiation, or occupational exposure.
- **Lifestyle factors:** None. Not related to smoking, diet, exercise, or alcohol.
- **Infectious agents:** Not applicable; CDD is non-infectious.

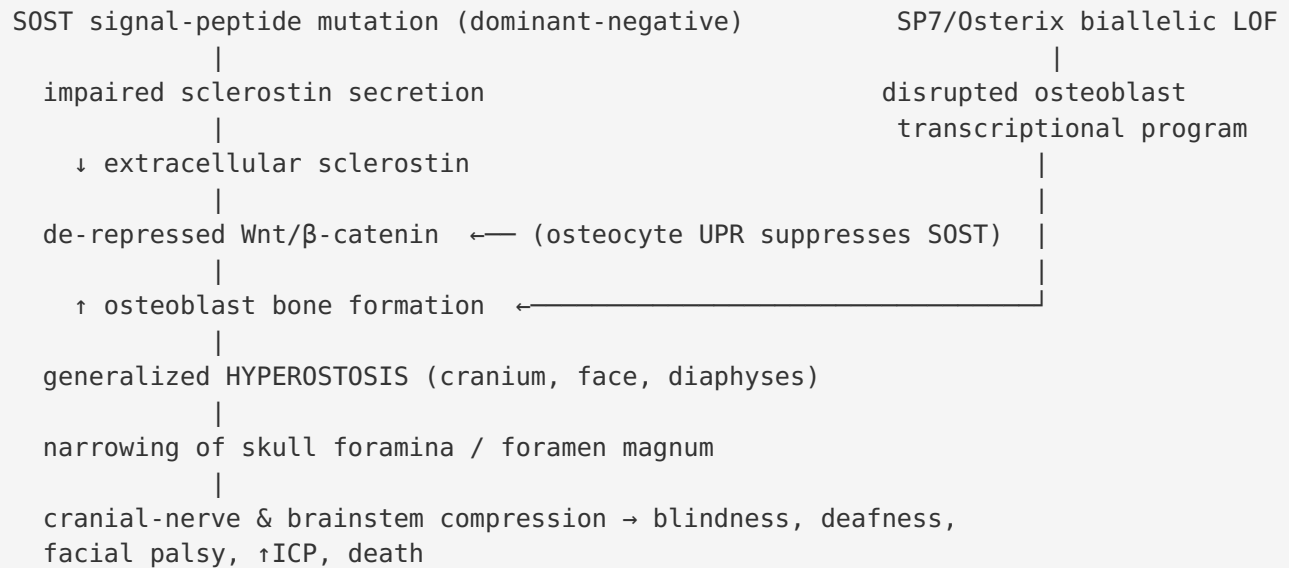
(The osteocyte UPR mechanism (Section 6) is an intrinsic cell-stress pathway, not an environmental exposure, though in principle it could be modulated pharmacologically.)

6. Mechanism / Pathophysiology

Ordered causal chain

1. A **heterozygous *SOST* signal-peptide mutation** (p.Val21Met / p.Val21Leu) is present in osteocytes → **impairs trafficking/secretion of sclerostin** (demonstrated in 293E cells) via a **dominant-negative** effect ([PMID: 21221996](#)).
2. Reduced secretion **leads to** decreased **extracellular sclerostin** available at the bone surface.
3. Because sclerostin normally **binds LRP4/5/6 to antagonize canonical Wnt/ β -catenin signaling**, its deficiency **results in de-repression (activation) of Wnt/ β -catenin signaling** in osteoblasts ([PMID: 19936252](#); [PMID: 35099616](#)).
4. Activated Wnt signaling **drives osteoblast commitment, differentiation, and enhanced periosteal/endosteal bone formation** → **generalized hyperostosis** ([PMID: 28973168](#)).
5. (*Inferred amplifier branch*) Osteocyte **ER stress / unfolded protein response (UPR)** can independently **delay osteocyte maturation and suppress *SOST* expression**, converging on the same "low sclerostin → active Wnt → hyperostosis" output ([PMID: 28973168](#)).
6. Sustained bone deposition **produces craniofacial + diaphyseal hyperostosis** that **narrows the skull foramina and the foramen magnum**.
7. Foraminal narrowing **compresses cranial nerves and the brainstem**, which **results in** optic atrophy/blindness, sensorineural + conductive hearing loss, facial palsy, raised intracranial pressure, and death ([PMID: 8827383](#)).

Convergent recessive branch: biallelic loss of *SP7/Osterix* disrupts osteoblast transcriptional programming and **leads to** an overlapping sclerotic skeletal phenotype ([PMID: 36436818](#); [PMID: 37918503](#)).



Detail by category

- **Molecular pathways:** Canonical **Wnt/β-catenin** signaling is the central pathway; sclerostin is its physiologic osteocyte-derived antagonist acting at **LRP4/5/6** co-receptors. Sclerostin also modulates **BMP** signaling ([PMID: 19936252](#)).
- **Cellular processes:** Osteoblast commitment, differentiation, matrix formation/mineralization; osteocyte maturation. GO suggestions: **GO:0001649** (osteoblast differentiation), **GO:0060348** (bone development), **GO:0045668** (negative regulation of osteoblast differentiation — normally exerted by sclerostin), **GO:0016055** (Wnt signaling pathway), **GO:0030282** (bone mineralization), **GO:0030968** (endoplasmic reticulum unfolded protein response).
- **Protein dysfunction:** Sclerostin ([UniProt Q9BQB4](#)) is a secreted glycoprotein; the signal-peptide mutation impairs its **secretion**, reducing the extracellular pool — a functional loss at the tissue level achieved through a dominant-negative cellular mechanism.
- **Metabolic changes:** No characteristic systemic metabolic derangement in CDD. (Osteocyte bioenergetics/PPARG links to sclerostin exist in the broader literature but are not CDD-specific.)
- **Immune system involvement:** None; CDD is not an immunologic or inflammatory disease.
- **Tissue-damage mechanisms:** Injury is **mechanical/compressive** — bony encroachment on neural foramina causing compression neuropathy of cranial nerves and brainstem, plus raised ICP — rather than oxidative, fibrotic, or necrotic.

- **Biochemical abnormalities:** Deficiency of functional extracellular sclerostin at the osteocyte–osteoblast interface.
- **Epigenetic changes:** Not established for CDD.
- **Molecular profiling:** No human transcriptomic/proteomic/metabolomic CDD datasets; mechanistic data derive from mouse models and in vitro assays.

Cell types involved (CL suggestions): osteocyte (CL:0000137), osteoblast (CL:0000062), osteoprogenitor/mesenchymal stem cell (CL:0000134). **Anatomical/UBERON:** cranium/skull (UBERON:0003128 / UBERON:0000209), facial bone, diaphysis of long bone (UBERON:0004769), periosteum (UBERON:0002515), foramen magnum.

7. Anatomical Structures Affected

Organ level. - *Primary:* **Bone** — skull/calvaria, facial bones, mandible; long-bone diaphyses; clavicles, ribs, axial skeleton (macrocephaly; near-4-cm calvarial thickness; PMID: 8827383; PMID: 14564212). - *Secondary:* **Nervous system** — cranial nerves (II optic, VII facial, VIII vestibulocochlear), brainstem, and brain, through foraminal narrowing and raised ICP (PMID: 8827383); **special-sense organs** (visual and auditory systems). - *Body systems:* skeletal (primary); nervous and sensory (secondary).

Tissue and cell level. Connective tissue — bone. Cellular effectors: **osteoblasts** (excessive bone formation) driven by loss of the **osteocyte**-derived sclerostin brake (CL:0000062, CL:0000137).

Subcellular level. Endoplasmic reticulum / secretory pathway — the signal-peptide mutation impairs ER-to-extracellular trafficking of sclerostin; osteocyte **ER stress/UPR** is implicated as an amplifier. GO cellular-component suggestions: **GO:0005783** (endoplasmic reticulum), **GO:0005576** (extracellular region), **GO:0005615** (extracellular space).

Localization / lateralization. Bilateral and symmetric skeletal involvement. Temporal-bone CT in the *SOST* hyperostosis spectrum shows "*diffuse osteosclerosis affecting the bilateral ossicular chains and internal auditory meatus, as well as stenosis of the bilateral internal auditory meatus*" (PMID: 40605263).

8. Temporal Development

- **Onset:** Early childhood; congenital predisposition with clinical manifestation in the first years of life. Reported probands are children (Kim's two patients; a 10-year-old with hearing loss, [PMID: 31132523](#); a patient requiring decompression at age 12, [PMID: 8827383](#)). Rare milder cases are diagnosed in adulthood (56-year-old woman, [PMID: 14564212](#)).
 - **Onset pattern:** insidious, chronic.
 - **Progression:** relentless and progressive — *"progressive encroachment of the craniofacial foramina and brain by the relentless deposition of bone"* ([PMID: 8827383](#)). No spontaneous remission.
 - **Disease course:** chronic, lifelong, progressive; not episodic or relapsing-remitting.
 - **Critical periods:** Childhood and adolescence — the window in which bony overgrowth compromises neural foramina and interventions (decompression) can preserve vision, hearing, and life.
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9. Inheritance and Population

Epidemiology. Extremely rare — described as *"extremely rare"* ([PMID: 21221996](#)) and *"a rare, sporadic form of craniotubular bone dysplasia"* ([PMID: 8827383](#)). Orphanet lists an estimated prevalence $<1/1,000,000$; fewer than ~two dozen cases are reported worldwide. Precise incidence/prevalence figures cannot be reliably estimated.

Genetic etiology. - **Inheritance: Autosomal dominant** (*SOST* signal-peptide mutations), typically de novo/sporadic; **autosomal recessive** (biallelic *SP7*/Osterix). - **Penetrance / expressivity:** Dominant *SOST* cases are severe and appear highly penetrant; expressivity is variable across the *SOST* hyperostosis spectrum (severe classic CDD to milder adult-diagnosed cases). - **Genetic anticipation:** Not applicable (no repeat-expansion mechanism). - **Germline mosaicism / founder effects / consanguinity:** No documented founder effect for CDD; consanguinity is relevant to the recessive *SP7* form. (By contrast, the related recessive disorder **sclerosteosis** shows a well-known Afrikaner founder effect: minimum prevalence $\sim 1/75,000$, gene frequency ~ 0.0035 , [PMID: 187366](#) — but this is a distinct disease.) - **Carrier frequency:** Not applicable for the dominant form; not established for recessive *SP7*.

Population demographics. No ethnic predilection established for CDD; cases are geographically scattered. Sex ratio approximately equal (autosomal inheritance); reported in both sexes, with pregnancies managed in an affected woman (PMID: 1987972). Age distribution skews to pediatric diagnosis.

10. Diagnostics

Clinical/imaging (the diagnostic cornerstone). - **Radiography/CT:** massive skull and facial-bone hyperostosis/sclerosis with facial distortion and macrocephaly; long-bone diaphyseal endostosis with undertubulation (loss of normal metaphyseal modeling); sclerotic clavicles, ribs, and axial skeleton (PMID: 8827383; PMID: 1987972; PMID: 14564212). Janssens et al.: *"marked sclerosis and hyperostosis of the skull bones is present resulting in macrocephaly. Most tubular bones of the limbs, as well as the clavicles, are affected by sclerosis"* (PMID: 14564212). - **Temporal-bone CT (SOST spectrum):** *"diffuse osteosclerosis affecting the bilateral ossicular chains and internal auditory meatus, as well as stenosis of the bilateral internal auditory meatus"* (PMID: 40605263). - **MRI:** skull thickening with loss of the diploic marrow signal. - **Laboratory tests / biomarkers:** No specific diagnostic blood/urine biomarker. Serum **sclerostin** could in principle be low; bone-turnover markers reflect bone formation but are non-specific. No validated CDD biomarker exists. - **Audiometry / ophthalmology:** to detect and monitor cranial-nerve compression (hearing loss, optic atrophy).

Genetic testing (confirmatory). - Recommended approach: targeted **single-gene SOST sequencing** for suspected dominant CDD; **WES/WGS** or a **sclerosing bone dysplasia gene panel** when the phenotype overlaps other craniotubular disorders, which also captures recessive **SP7/Osterix** (PMID: 21221996; PMID: 36436818). - CMA / karyotype / FISH / mtDNA / repeat-expansion testing are **not** indicated (single-gene, non-structural, non-mitochondrial, non-repeat disorder).

Clinical criteria / differential diagnosis. Diagnosis rests on the characteristic craniofacial + diaphyseal radiographic pattern plus molecular confirmation. CDD must be differentiated from other craniotubular hyperostoses:

Condition	Gene	Inheritance	Distinguishing features vs CDD
Craniodiaphyseal dysplasia	<i>SOST</i> (signal peptide) / <i>SP7</i>	AD / AR	Most severe; massive facial hyperostosis; no syndactyly
Sclerosteosis	<i>SOST</i> (LOF)	AR	Syndactyly of 2nd/3rd fingers , gigantism (PMID: 1259284; PMID: 187366)
Van Buchem disease	<i>SOST</i> (regulatory)	AR	Milder; no syndactyly
Craniometaphyseal dysplasia	<i>ANKH/GJA1</i>	AD/AR	Metaphyseal (not diaphyseal) flaring; can be misdiagnosed as CDD (PMID: 40639871)
Camurati-Engelmann disease	<i>TGFB1</i>	AD	Diaphyseal dysplasia; <i>TGFB1</i> excluded in a CDD case (PMID: 14564212)

The three *SOST*-related craniotubular hyperostoses are grouped mechanistically: *"Loss of sclerostin gene function is related to 3 different craniotubular hyperostosis processes: sclerosteosis, craniodiaphyseal dysplasia, and van Buchem disease"* (PMID: 29264888). The recessive disorder sclerosteosis is set apart by additional limb findings: *"Sclerosteosis is a unique autosomal recessive condition in which skeletal overgrowth is associated with syndactyly and digital malformation"* (PMID: 1259284).

Screening. No population newborn or carrier screening (ultra-rare, mostly de novo dominant). **Cascade/prenatal testing** is possible in families with a known variant.

11. Outcome / Prognosis

- **Survival / mortality:** Guarded. The natural course leads to *"compression of cranial nerves, the foramen magnum, and intracranial contents [that] commonly leads to blindness, loss of hearing, and death"* (PMID: 8827383). No formal 5-/10-year survival statistics exist owing to rarity; premature death from brainstem/foramen-magnum compression or raised ICP is a recognized outcome.

- **Morbidity / disability:** High and progressive — blindness, deafness, facial palsy, chronic headache, facial disfigurement, neurological impairment. Substantial lifelong disability.
- **Quality of life:** Severely affected across sensory, functional, and psychosocial domains; no disease-specific QoL instruments applied.
- **Complications:** Cranial-nerve palsies, raised intracranial pressure, seizures, and (in the recessive *SP7* form) recurrent fractures (PMID: 37918503).
- **Recovery potential:** No spontaneous recovery; surgical decompression can preserve function and life but does not halt the underlying bone deposition.
- **Prognostic factors:** Severity and rate of foraminal narrowing; timeliness of decompressive surgery; degree of cranial-nerve involvement at presentation. No molecular prognostic biomarker validated.

12. Treatment

No approved disease-modifying pharmacotherapy exists. Management is symptomatic, surgical, and rehabilitative.

Surgical / interventional (mainstay). - **Decompressive craniectomy** for raised ICP: *"Significant brain compression with signs and symptoms of increased intracranial pressure was managed successfully with decompressing craniectomy at age 12 years, enlarging the anterior and middle fossae. Calvarial thickness measured nearly 4 cm"* (PMID: 8827383). - **Staged craniofacial / mandibular reduction osteoplasty** (recontouring) for deformity and foraminal decompression (PMID: 8827383). (NCIT: cranial decompression; craniofacial reconstructive surgery.) - Decompression of specific neural foramina (e.g., optic canal, internal auditory meatus) to preserve vision and hearing.

Supportive / rehabilitative. - **Bone-anchored hearing aid (BAHA)** for hearing rehabilitation: *"we describe the first case of craniodiaphysial dysplasia rehabilitated with Bone-Anchored Hearing Aid, despite the concerns inherent to the involvement of the skull bone that characterizes the disease"* (PMID: 31132523). (NCIT: hearing aid; auditory rehabilitation.) - Ophthalmologic and audiologic surveillance; symptomatic management of headache and seizures.

Pharmacotherapy / advanced therapeutics. None approved. The sclerostin–Wnt axis is **pharmacologically validated in the opposite direction:** the anti-sclerostin antibody **romosozumab** is bone-anabolic — *"Romosozumab, a sclerostin inhibitor with both anabolic and antiresorptive effects"* (PMID: 42761068) — increasing spine and hip BMD in osteoporosis

(lumbar spine $+14.47 \pm 8.74\%$, $p < 0.001$; total hip $+4.15 \pm 5.92\%$, $p < 0.001$; [PMID: 42417976](#)). This confirms that *lowering* sclerostin drives bone gain (the CDD direction) but implies there is **no safe pro-sclerostin ("bone-quieting") therapy** currently available. A theoretical CDD therapy would need to *restore* Wnt inhibition (e.g., sclerostin replacement/mimetic or LRP6 modulation) — the reverse of osteoporosis drug development.

Experimental / preclinical leads. - **Chemical chaperone sodium 4-phenylbutyrate (4-PBA):** suppressed hyperostosis in the osteocyte-UPR mouse model consistent with CDD — *"A clear relationship between the activation of the unfolded protein response was established and the onset of hyperostosis that can be suppressed with a chemical chaperone, sodium 4-phenobutyrate (4-PBA)"* ([PMID: 28973168](#)). - **LRP4–sclerostin interface modulation:** competitive blocking studied for anabolism ([PMID: 35099616](#)) — conceptually reversible for CDD.

Personalized/genotype-guided care. Distinguishing dominant *SOST* from recessive *SP7* CDD informs recurrence-risk counseling and prognosis but does not yet change pharmacologic management.

13. Prevention

- **Primary prevention:** Not possible — a genetic disorder, largely de novo. No modifiable risk factors.
- **Secondary prevention:** Early diagnosis (radiographic + molecular) and **surveillance** of vision, hearing, and ICP to time decompressive surgery before irreversible cranial-nerve damage — the principal opportunity for prevention of morbidity.
- **Tertiary prevention:** Surgical decompression and reduction osteoplasty to prevent blindness, deafness, and death from foraminal/brainstem compression; hearing rehabilitation.
- **Immunization / public-health / environmental interventions:** Not applicable.
- **Genetic counseling:** Central. For dominant *SOST* CDD, counsel on de novo occurrence and 50% transmission risk from an affected parent; for recessive *SP7* CDD, counsel on 25% recurrence and consanguinity. **Prenatal / preimplantation genetic testing** is feasible when the familial variant is known.

14. Other Species / Natural Disease

- **Taxonomy:** No naturally occurring companion-animal or wildlife equivalent of CDD is documented in OMIA. Disease knowledge derives from engineered laboratory models (Section 15).
 - **Orthologous genes:** *Sost* (mouse, **NCBI Gene 74499**; *Mus musculus*, **NCBI:txid10090**); *Sp7*/Osterix (mouse). Both are conserved and functionally validated (below).
 - **Comparative biology:** *Sost*-deficient mice recapitulate the high-bone-mass consequence of sclerostin loss, and "*Sost/SOST deficiency induces lifelong bone gain in mice and humans*" ([PMID: 23901037](#)) — demonstrating strong evolutionary conservation of the sclerostin–Wnt bone-formation mechanism.
 - **Transmission / zoonosis:** Not applicable (non-infectious genetic disease).
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15. Model Organisms

Mammalian (mouse) models are the principal system.

Model	Type	Key phenotype	Relevance to CDD	Evidence
<i>Sost</i> ^{-/-} (global KO)	Knockout	High bone mass via increased bone formation; elevated femoral-neck BV/TV	Models the sclerostin-deficiency → high-bone-mass output	PMID: 33339872; PMID: 31437568
<i>Sost/SOST</i> deficiency (mouse + human)	Genetic	Lifelong bone gain	Cross-species validation of mechanism	PMID: 23901037
AAV8-Sp7-Cre postnatal <i>Sost</i> KO	Conditional/somatic	Increased bone anabolism in adults; decreased canalicular density	Confirms postnatal role of <i>Sost</i> in bone homeostasis	PMID: 36462771
Osteocyte-UPR transgenic	Transgenic (gain of ER stress)	Generalized hyperostosis; SOST suppressed; Wnt activated; rescued by 4-PBA	Phenotype " <i>highly consistent with craniodiaphyseal dysplasia</i> " — closest CDD-specific model	PMID: 28973168

Phenotype recapitulation. *Sost*-loss models faithfully reproduce the **high-bone-mass / hyperostosis** endpoint central to CDD ("*Loss-of-function mutations in the Sost gene lead to high bone mass phenotypes*", PMID: 33339872), and the osteocyte-UPR transgenic reproduces **generalized hyperostosis explicitly likened to CDD** with a druggable rescue: "*As the phenotype is highly consistent with craniodiaphyseal dysplasia (CDD; OMIM 122860), we propose activation of the UPR could be part of the disease mechanism for CDD patients*" (PMID: 28973168).

Limitations. No mouse carries the exact human *SOST* signal-peptide (p.Val21Met/Leu) dominant-negative allele; existing KOs model loss of function rather than the dominant-negative secretion defect. Murine models capture bone overgrowth but incompletely reproduce the human craniofacial "leontiasis ossea" and cranial-nerve compression syndrome. A caution flag from the literature: *Sost* haploinsufficiency combined with glucocorticoid excess produced lethal cardiac tamponade in mice (PMID: 30664862), highlighting potential off-target cardiovascular effects of sclerostin-axis manipulation.

Resources: MGI (mouse *Sost*, *Sp7*); IMPC/KOMP for conditional alleles.

Mechanistic Model / Interpretation

CDD is best understood as a **"loss-of-the-brake" bone disease**. Sclerostin is the physiologic osteocyte-derived antagonist that restrains Wnt-driven osteoblast bone formation by binding LRP4/5/6. In dominant CDD, a signal-peptide mutation blocks sclerostin from leaving the cell (a dominant-negative secretion defect), so the extracellular brake is lost, canonical Wnt/ β -catenin signaling runs unchecked, and osteoblasts deposit bone relentlessly across the skull, face, and diaphyses. Because the skull is a closed compartment perforated by fixed foramina, the pathology's clinical severity comes not from any metabolic toxicity but from **mechanical geometry**: bone fills the foramina and cranial cavity, strangling cranial nerves II/VII/VIII and the brainstem. The osteocyte UPR provides a plausible intracellular amplifier that further suppresses *SOST* and can, on its own, generate a CDD-like phenotype in mice. A parallel recessive route through *SP7*/Osterix loss reaches an overlapping sclerotic phenotype by disrupting the osteoblast transcriptional program upstream/parallel to the sclerostin–Wnt node.

The therapeutic corollary is striking and well-supported: pharmaceutical companies deliberately **inhibit** sclerostin (romosozumab) to *build* bone in osteoporosis. CDD is essentially the endogenous, lifelong version of that intervention. This confirms the causal direction beyond doubt but also explains why no drug exists for CDD — the field has optimized tools to *lower* sclerostin, whereas CDD needs the opposite (restore Wnt inhibition), a direction with no approved agent and the added hazard, seen in mice, of cardiovascular effects from sclerostin-axis perturbation.

Evidence Base

PMID	Title (abbrev.)	Contribution
21221996	<i>SOST signal-peptide mutations in AD CDD</i>	Landmark — identifies causal p.Val21Met/Leu; demonstrates reduced secretion; establishes dominant-negative mechanism
8827383	<i>Reduction osteoplasty for CDD</i>	Core clinical phenotype, progression, cranial-nerve compression, surgical decompression
28973168	<i>Osteocyte UPR causes CDD-consistent hyperostosis</i>	UPR→SOST suppression→Wnt→hyperostosis; 4-PBA rescue; CDD-like mouse model
19936252	<i>Lrp4, receptor for sclerostin</i>	Sclerostin binds LRP5/6 and inhibits Wnt — normal function whose loss drives CDD
35099616	<i>Blocking LRP4–sclerostin interface</i>	Confirms LRP4/6–sclerostin mechanism; reversible-modulation concept
36436818	<i>Recessive CDD from SP7/Osterix</i>	Establishes genetic heterogeneity (recessive form)
37918503	<i>SP7-related bone disorder follow-up</i>	Biallelic SP7 LOF variants; CDD-overlapping sclerotic dysplasia
29264888	<i>Sclerostin mechanisms/disorders</i>	Groups CDD with sclerosteosis & van Buchem (SOST spectrum)
1259284 / 187366	<i>Sclerosteosis clinical features</i>	Differential diagnosis: recessive, syndactyly, founder effect
40639871	<i>CMD misdiagnosed as CDD</i>	Differential: craniometaphyseal dysplasia
14564212	<i>Mild CDD case</i>	Milder/adult phenotype; excluded TGFB1/LRP5; radiographic features
40605263	<i>SOST LOF, sclerosteosis-1</i>	Temporal-bone CT findings in SOST hyperostosis spectrum
31132523	<i>BAHA in CDD</i>	Hearing rehabilitation option
1987972	<i>Pregnancy in CDD</i>	Raised ICP, cranial-nerve palsies, seizures, long-bone modeling defects

PMID	Title (abbrev.)	Contribution
23901037	<i>Reversing SOST-deficiency disorders</i>	Cross-species: SOST deficiency → lifelong bone gain
33339872 / 31437568 / 36462771	<i>Sost KO mouse models</i>	High-bone-mass phenotype validating the mechanism
42761068 / 42417976	<i>Romosozumab</i>	Anti-sclerostin antibody is bone-anabolic — validates CDD causal direction
30664862	<i>Sost haploinsufficiency → cardiac tamponade</i>	Safety caution for sclerostin-axis manipulation

Evidence-source mix: human clinical case reports/series (phenotype, surgery, genetics), in vitro secretion assay (293E cells), and model-organism (mouse) studies. No large human cohorts, omics datasets, or randomized trials exist for CDD itself.

Limitations and Knowledge Gaps

1. **Tiny evidence base.** Fewer than ~two dozen reported CDD cases; almost all knowledge is from case reports. Prevalence, incidence, penetrance, and QoL are estimated qualitatively, not measured.
2. **Genotype–phenotype spectrum incompletely defined.** Only two dominant *SOST* signal-peptide alleles and a few recessive *SP7* variants are described; the full mutational spectrum and modifiers are unknown.
3. **Dominant-negative mechanism partly inferred.** Reduced secretion is demonstrated in a cell line; how mutant sclerostin dominantly interferes with wild-type protein at the tissue level is not fully resolved. The UPR amplifier is a plausible but not yet human-confirmed contributor.
4. **No faithful animal model of the human allele.** Existing mice model loss-of-function or ER stress, not the specific dominant-negative signal-peptide defect, and do not fully reproduce craniofacial/cranial-nerve pathology.

5. **No disease-modifying therapy or trials.** Management is entirely symptomatic/surgical; the required "pro-sclerostin/Wnt-inhibition" direction has no approved drug and carries potential cardiovascular risk.
 6. **No CDD-specific omics.** No transcriptomic, proteomic, metabolomic, or epigenomic patient datasets exist.
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Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse** carrying the human *SOST* p.Val21Met (or p.Val21Leu) allele to model the dominant-negative secretion defect and test whether it reproduces craniofacial hyperostosis and cranial-nerve compression.
 2. **Test sclerostin-restoring / Wnt-inhibiting strategies** preclinically — recombinant/engineered sclerostin mimetics, LRP6 antagonism, or DKK1 pathway modulation — to establish proof-of-concept for slowing bone deposition, monitoring for the cardiovascular signal flagged in [PMID: 30664862](#).
 3. **Evaluate chemical chaperones (4-PBA)** and broader UPR modulators in CDD-relevant models, building on [PMID: 28973168](#), toward a repurposing rationale.
 4. **Establish an international CDD registry / natural-history study** pooling *SOST* and *SP7* cases to quantify onset, progression rate, cranial-nerve outcomes, surgical timing/efficacy, and survival.
 5. **Standardize molecular diagnosis** via a sclerosing bone dysplasia gene panel (including *SOST*, *SP7*, *ANKH*, *TGFB1*, *LRP5*) with functional secretion assays for novel *SOST* variants.
 6. **Patient-derived iPSC osteocyte/organoid models** to study the dominant-negative secretion defect and UPR in a human genetic background and to screen candidate therapeutics.
 7. **Prospective surveillance protocols** (serial vision, audiometry, ICP/imaging) to define optimal timing of decompressive surgery — the current best lever on morbidity and mortality.
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Report compiled from 12 confirmed findings across 33 reviewed papers over 5 investigation iterations. Evidence types: human clinical (case reports/series), in vitro (secretion assays), and model-organism (mouse). CDD remains an ultra-rare disorder where mechanistic understanding substantially outpaces therapeutic options.

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